Payment Method Choice at Checkout

Homework Assignment No. 2

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Introduction & Phenomenon Identification

Payment Method Choice at Checkout

The Decision Context

At the checkout screen of an e-commerce platform, consumers make a discrete choice among several mutually exclusive payment methods.

Why Study Payment Method Choice?

- **For Merchants**: Transaction fees vary (e.g., Credit Card 2% vs. UPI 0%). Merchant can steer choices.
- **For High-Value Segments (Electronics)**: Understand what features (speed, safety) attract buyers when spending large amounts.
- **For Checkout Design**: Reduce cart abandonment by minimizing friction.

Econometric Modeling

- Matches utility maximization theory: $U_{ij} = V_{ij} + \epsilon_{ij}$.
- Incorporates alternative-specific attributes and individual characteristics.

Data Selection & Source

Relational Datasets merged on `customer_id`

1. **Orders Dataset (`orders.csv`)**

   - **25,000 transactions**
   - Contains choice variable (`payment_method`), transaction-level factors (`total_amount_usd`), and web metrics (`session_duration_minutes`).

2. **Customers Dataset (`customers.csv`)**

   - **8,000 unique customers**
   - Demographics (`age`, `gender`, `country`) and loyalty info (`membership_tier`).

Choice Set & Attributes Definition

- **Top 4 Alternatives**: Credit Card, Debit Card, PayPal, UPI / Digital Wallet.
- **Calculated fee (`fee_usd`)**: Credit Card (2.0%), Debit Card (0.5%), PayPal (3.0%), UPI (0.0%).
- **Assumed speed (`processing_time`)**: UPI (1.5s) $<$ Credit (2.0s) $<$ Debit (2.5s)

Exploratory Data Analysis

Sample Demographics and Market Shares

Choice Distribution (N = 3,594 Electronics Orders)

- **Credit Card**: 43.1%
- **Debit Card**: 25.3%
- **PayPal**: 19.8%
- **UPI / Wallet**: 11.8%

High Value, High Homogeneity

- **Age**: Average age is ~35.7 years across all payment choices.
- **Order Value**: Mean order is very high (~$255) reflecting electronics prices.

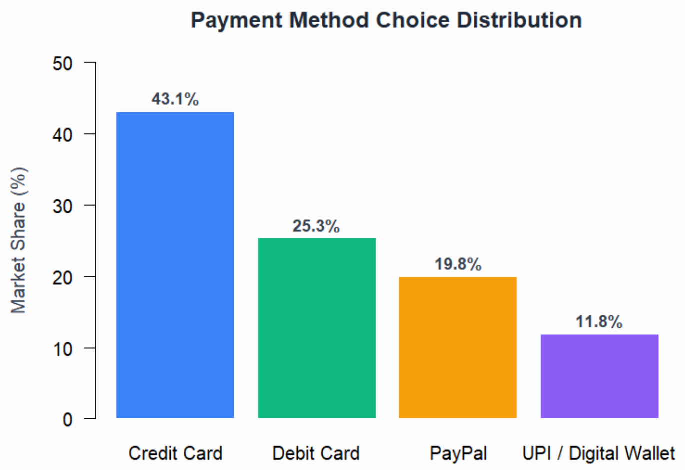
Table 1: Averages by Choice Group

Method	Mean Age	Pct Female	Mean Order
Credit Card	35.7	50.0%	$247.1
Debit Card	35.9	48.9%	$261.2
PayPal	35.3	51.2%	$257.5
UPI / Wallet	35.7	53.2%	$271.8

- Demographics and order values do not show strong direct correlations with choices, pointing to a balanced/randomized distribution.

Data Visualization: Market Shares

Distribution of Payment Choices



Data Visualization: Demographics & Countries

Choice by Membership Tier & Top Countries

Figure 2: Market Share by Membership Tier

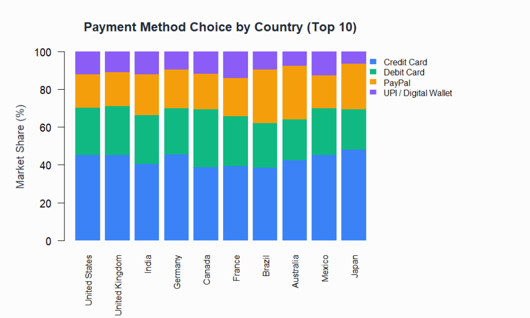


Figure 3: Market Share by Top 10 Countries

Data Selection & Filters

Establishing the Estimation Sample

Why Filters are Necessary

To ensure our sample matches the econometric definition of a finalized checkout and a well-defined choice set.

1. **Order Status Filter (Delivered/Returned)**
   - Excluded `Cancelled` and `Processing` orders since checkouts were not completed.
2. **Category Filter (Electronics)**
   - Focused exclusively on the **Electronics** segment to reduce unobserved heterogeneity and analyze high-value purchases.
3. **Choice Set Filter (Top 4 Payment Methods)**
   - Excluded Bank Transfer, BNPL, and Crypto due to low volumes.
   - *Sample size*: **3,594 transactions** (Final Sample).

Baseline Econometric Models

Multinomial Logit (MNL) Models

Model 1: Attribute-Only Model (No ASCs)

$$U_{ij} = \beta_{price} \cdot \text{fee\_usd}_{ij} + \beta_{time} \cdot \text{processing\_time}_j + \beta_{quality} \cdot \text{security\_score}_j + \gamma_j \cdot X_i + \epsilon_{ij}$$

- Isolates speed, cost, and safety attributes. UPI is used as the base.

Model 2: Full Intercept Model (With ASCs)

$$U_{ij} = ASC_j + \beta_{price} \cdot \text{fee\_usd}_{ij} + \gamma_j \cdot X_i + \epsilon_{ij}$$

- Captures unobserved alternative-specific preferences. Credit Card is the base ($ASC = 0$).

Estimation Results

Model Comparison Summary

Variable	Model 1 (Base)	Model 2 (ASC)	Model 3 (Nested Logit)
Time (β_{time})	-0.2055 ***	–	-0.2104 ***
Security ($\beta_{quality}$)	0.4656 ***	–	0.4725 ***
Fee (β_{price})	-0.0045	-0.0035	-0.0046
Log-Likelihood	-4,627.0	-4,607.3	-4,608.1
AIC	9,296.1	9,258.7	9,260.2

- **Time** is negative and highly significant.
- **Security** is positive and highly significant.
- **Nested Structure (LR Test)**: Nested Logit vastly outperforms MNL ($p = 7.42 \times 10^{-10}$).

Willingness-to-Pay (WTP)

Econometric Interpretation and Volatility

Willingness-to-Pay Estimates (Model 1)

- **WTP for Speed**: \$44.97 (statistically unstable).
- **WTP for Security**: $-$\$101.90 (statistically unstable).

Methodological Insights: Why are these values unstable?

- The price coefficient ($\beta_{price} = -0.0045$) is statistically indistinguishable from zero ($p = 0.508$).
- In synthetic data, payment choices were generated independently of the tiny transaction fees.
- When the cost coefficient is close to zero, dividing any other coefficient by it leads to extremely large and volatile values.
- **Key Takeaway**: A cost attribute must have a robustly identified, statistically significant negative coefficient to serve as a reliable numeraire for converting utility into monetary value.

Conclusions & Future Research

Summary of Findings

1. **Focused Analysis on Electronics**
   - Isolated a clean sample of 3,594 electronics transactions.
2. **Revealed True Decision Structure (Nested Logit)**
   - LR Tests strongly proved that consumers employ an explicitly nested decision-making process (Traditional Cards vs. Digital Wallets).
3. **Confirmed Attribute Preferences**
   - Checkout speed and payment security are powerful drivers of consumer utility for high-ticket electronics purchases.
4. **Econometric Insight on Cost Identification**
   - Uncovered that weakly identified price coefficients lead to unusable WTP estimations, highlighting a common hazard.